Introduction

In this tutorial, we will explore a modelling framework that refines the **traditional degree-day approach** for describing larval development in relation to temperature. This framework is implemented in the **Population package** and the companion **PopJSON language**, designed to make population modelling more transparent, flexible, and reproducible.

The approach is also described in recent publications Erguler et al. 2018, *F1000Research* and Erguler et al. 2022, *Scientific Reports*.

From Degree-Days to Data-Driven Development

Traditionally, growth is calculated by how many **degree-days** an organism accumulates above a baseline temperature.

For example:

- If larvae require at least 10 °C to grow, then spending one day at 11 °C counts as one degree-day.
- A day at 12 °C counts as two degree-days, and so on.

Instead of assuming a perfectly linear relationship between temperature and development, we allow for a complex dependencies, **species-specific, non-linear functions**, informed by **experimental data**. For instance,

- If larvae take 20 days to complete development at 20 °C, we assume they accumulate 1/20 of their development each day.
- At 30 °C, if development takes 5 days, then accumulation proceeds at 1/5 per day.

Our model relaxes the idea of a **fixed heat accumulation threshold**.

- Instead of requiring, for example, exactly 3,000 degree-days for development, we treat development time as a **distribution**.
- A population might develop around 30 days with some variability.
- This variability can be described using an **Erlang distribution** or other probabilistic approaches.

Finally, our model incorporates **multiple time-dependent processes** acting on a population. We assume that each process realises in sequence and dynamically structures the population into a set of pseudo-stages, each with a specific time distribution.

This tutorial will guide you through building such models step by step, using Population and PopJSON to put these concepts into practice.

In [1]:
```
# This is a wrapper that comes with the PopJSON library (written in NodeJS)
# It exists in $POPJSON_WPATH or node_modules/popjson/wrappers/
# Here, it is symbolically linked to the working directory.
#
# sys.call(sprintf("cp /opt/conda/lib/node_modules/popjson/wrappers/population.* ./")
#
source("population.R")
```

In [2]:
```
# This is a convenience script to parse PopJSON models into C (which implements the P
# and compile as a "dylib" for access from R
#
prepare <- function(filename) {
    out <- system(sprintf("popjson %s", filename), intern = TRUE)
    print(out)
    #
    if (exists("model") && inherits(model, "PopulationModel")) model$destroy()
    #
    model <- PopulationModel$new(sprintf("%s.dylib", filename))
    #
    return(model)
}
```

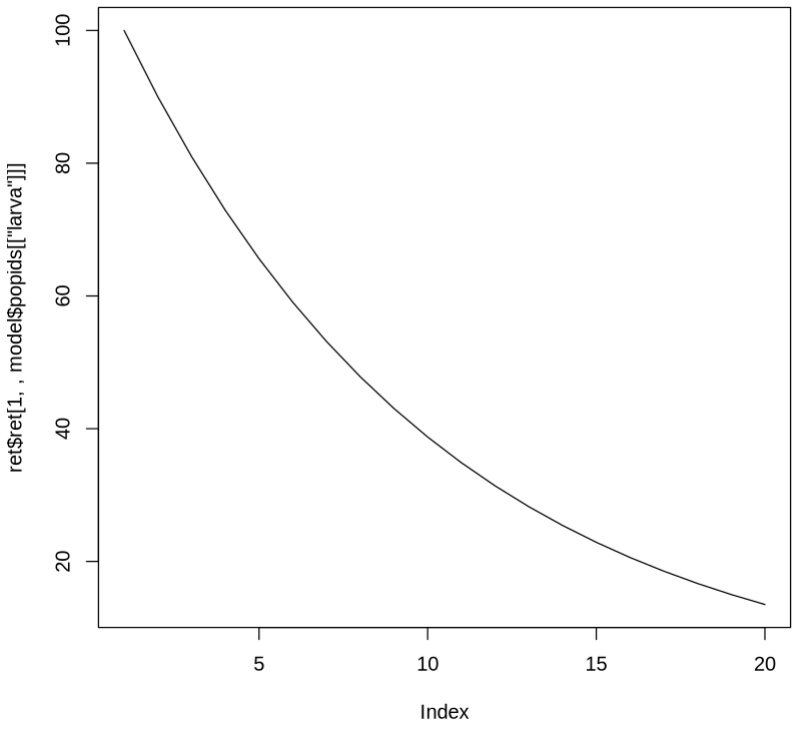
Let's use models/example.json to represent larval growth dynamics.

Please refer to **Population package** and **PopJSON language** for descriptions of what's possible.

In [5]:
```
model <- prepare("models/example")
```

```
[1] "Model translated to models/example.c"
[2] "Model file created: models/example.dylib"
Loaded model with: 1 pops, 0 pars, 0 ints, 0 envs.
```

In [6]:
```
ret <- model$sim(ftime = 20,
                 y0 = list(larva = 100.0))

plot(ret$ret[1, , model$popids[["larva"]]], t = "l")
```

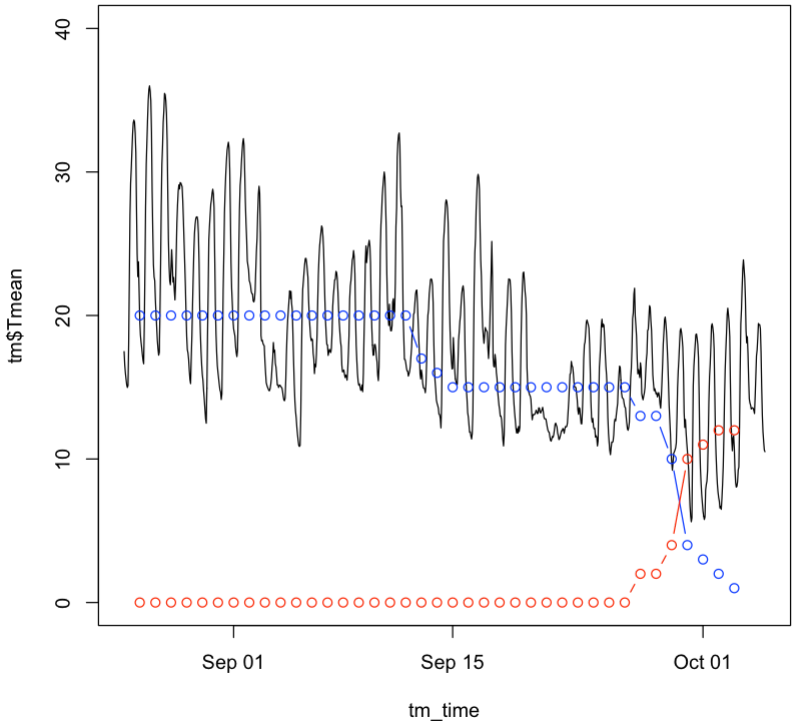


Semi-field experiments

Let's try investigating something more complex but realistic. In 2022, we published an analysis using the set of semi-field experiments on immature stage development of *Culex pipiens* biotype

molestus, performed by Dusan Petric's group. Please refer to Erguler et al. 2022, *Scientific Reports* for more information.

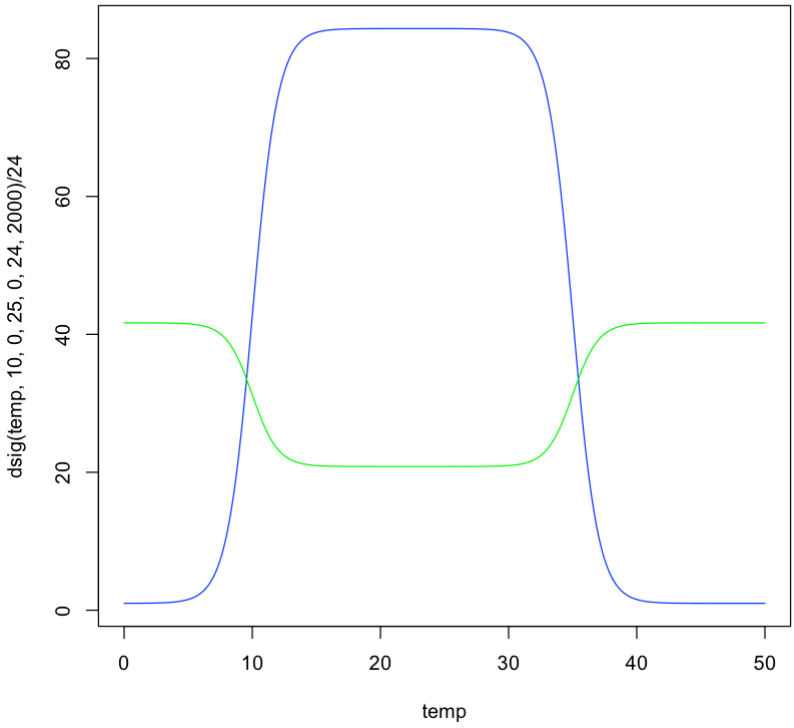
In [67]:
```
df <- read.csv("data/data_semifield.csv")
df_time <- as.POSIXct(df$Date, format="%d/%m/%Y")

tm <- read.csv("data/datalogger.csv")
tm_time <- as.POSIXct(tm$Time, format="%Y-%m-%d %H:%M:%S")

plot(tm_time,tm$Tmean,t='l',ylim=c(0,40))
lines(df_time,df$L,t='b',col='blue')
lines(df_time,df$P,t='b',col='red')
```



We propose an equation to link temperature to larva development and survival. The double sigmoidal is my favourite for its flexibility and ease of interpretation.

In [ ]:
```
dsig <-function(T,L,l,R,r,M,X) {
    return (((M) + ((X) / ((1.0 + exp(exp((l)) * ((L) - (T)))) * (1.0 + exp(exp((r))
}
```

In [6]:
```
temp <- seq(0, 50, length.out = 1000)

plot(temp, dsig(temp,10,0,25,0,24,2000) / 24, t='l', col='blue')
lines(temp, dsig(temp,10,0,25,0,1000,-500) / 24, t='l', col='green')
```

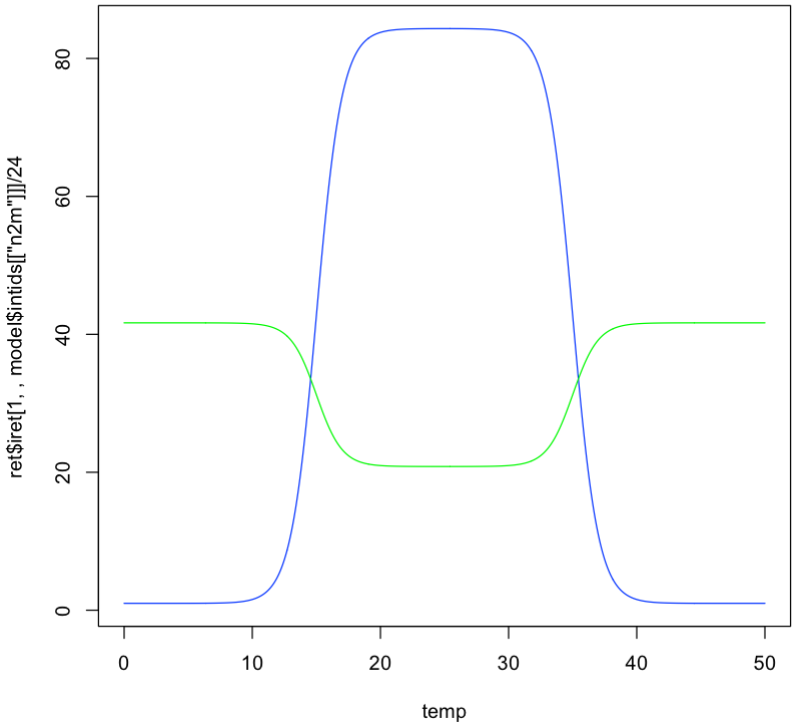


Let's go over the rest one step at a time, together...

In [56]:
```
model <- prepare("models/model_larva")
```

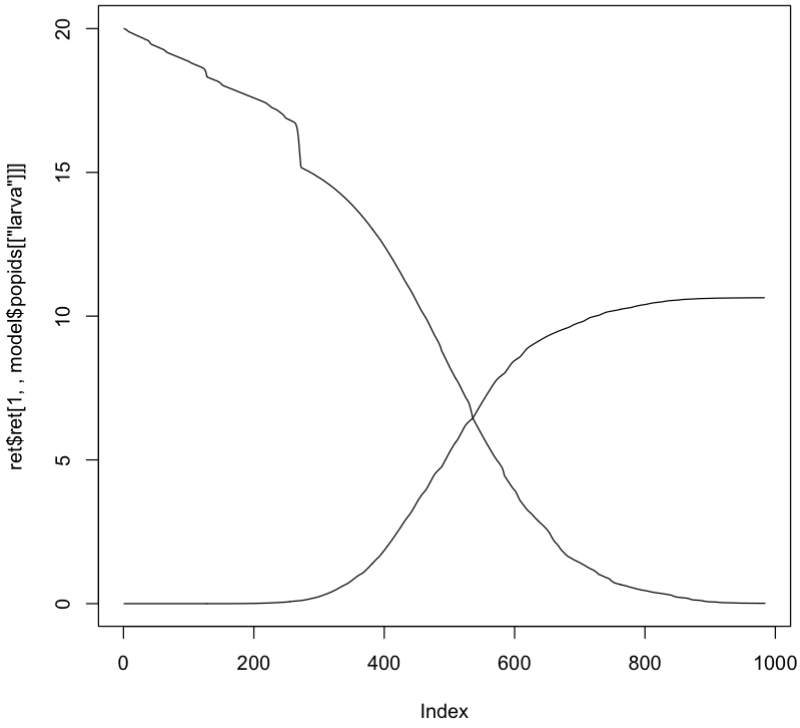
```
[1] "Reading model file: models/model_larva.json"
[2] "Model translated to models/model_larva.c"
[3] "Model file created: models/model_larva.dylib"
Loaded model with: 1 pops, 12 pars, 4 ints, 1 envs.
```

In [57]:
```
temp <- seq(0,50,length.out=1000)

param <- model$param

ret <- model$sim(
  ftime = length(temp)+1,
  envir = list(
    temp = temp
  ),
  pr = param,
  rep = -1
)

plot(temp, ret$iret[1, , model$intids[["n2m"]]] / 24.0, t='l', col='blue')
lines(temp, ret$iret[1, , model$intids[["d2m"]]] / 24.0, t='l', col='green')
```

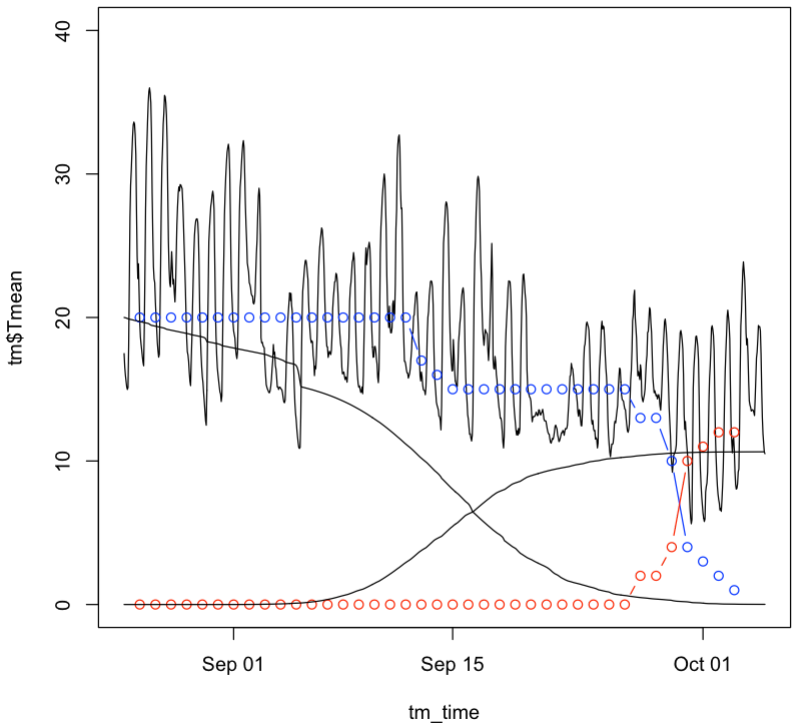


In [58]:
```
temp <- tm$Tmean

param <- model$param

ret <- model$sim(
  ftime = length(temp),
  envir = list(
    temp = temp
  ),
  pr = param,
  y0 = list(
    larva = 20.0
  )
)
```

In [59]:
```
plot(ret$ret[1, , model$popids[["larva"]]], t = "l")
lines(cumsum(ret$iret[1, , model$intids[["larva_to_pupa"]]]), t = "l")
```



In [60]:
```
plot(tm_time,tm$Tmean,t='l',ylim=c(0,40))
lines(df_time,df$L,t='b',col='blue')
lines(df_time,df$P,t='b',col='red')
lines(tm_time,ret$ret[1, , model$popids[["larva"]]], t = "l")
lines(tm_time[-length(tm_time)],cumsum(ret$iret[1, , model$intids[["larva_to_pupa"]]]
```



In [61]:
```
obs <- df
obs['hour'] <- (1:nrow(obs))*12

simulate_system <- function(param) {
  temp <- tm$Tmean
  hours <- length(temp)

  ret <- model$sim(
```

```
    ftime = hours,
    envir = list(
      temp = temp
    ),
    pr = param,
    y0 = list(
      larva = 20.0
    )
  )

  return(data.frame(hour = (1:hours)-1,
                    larva = ret$ret[1, , model$popids[["larva"]]],
                    pupa = c(cumsum(ret$iret[1, , model$intids[["larva_to_pupa"]]])),
}

objective_function <- function(param) {
  sim <- simulate_system(param)
  # Extract simulation only at obs times
  sim_at_obs <- sim[tm_time %in% df_time, ]
  # sum of squared errors (larva + pupa)
  SSE <- sum((obs$L - sim_at_obs$larva)^2 + (obs$P - sim_at_obs$pupa)^2)
  return(SSE)
}
```

In [62]:
```
objective_function(param)
```

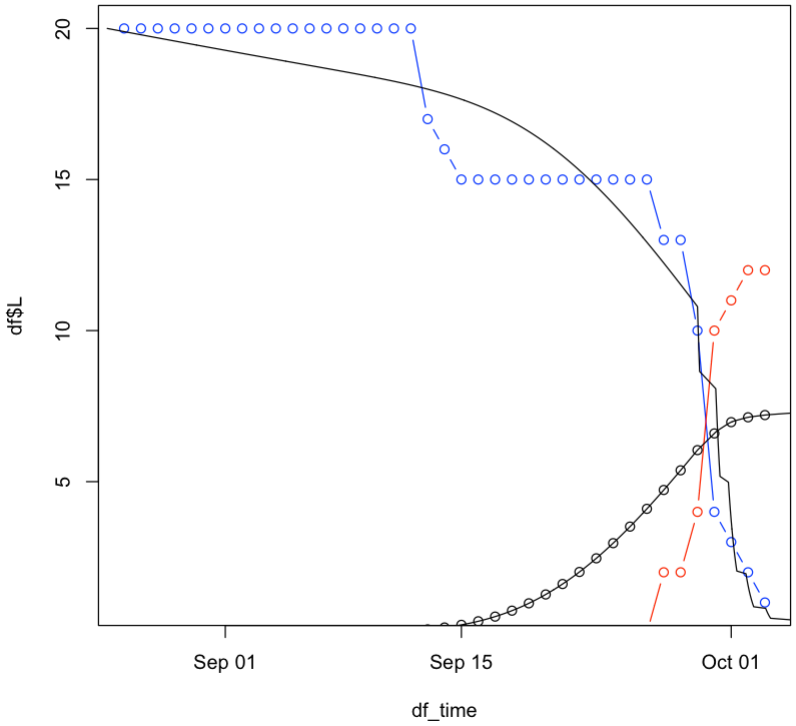
3722.04044709227

In [63]:
```
lower_bounds <- model$parmin
upper_bounds <- model$parmax

fit <- optim(
  par = model$param,
  fn = objective_function,
  method = "L-BFGS-B",
  lower = lower_bounds,
  upper = upper_bounds
)

print(fit)
```

```
$par
 [1]    9.892222    5.983674   50.000000    6.000000   14.029705 4550.024496
 [7]   50.000000    6.000000  -10.000000   -6.000000  465.430554  946.066874

$value
[1] 232.7981

$counts
function gradient
     137      137

$convergence
[1] 0

$message
[1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"
```

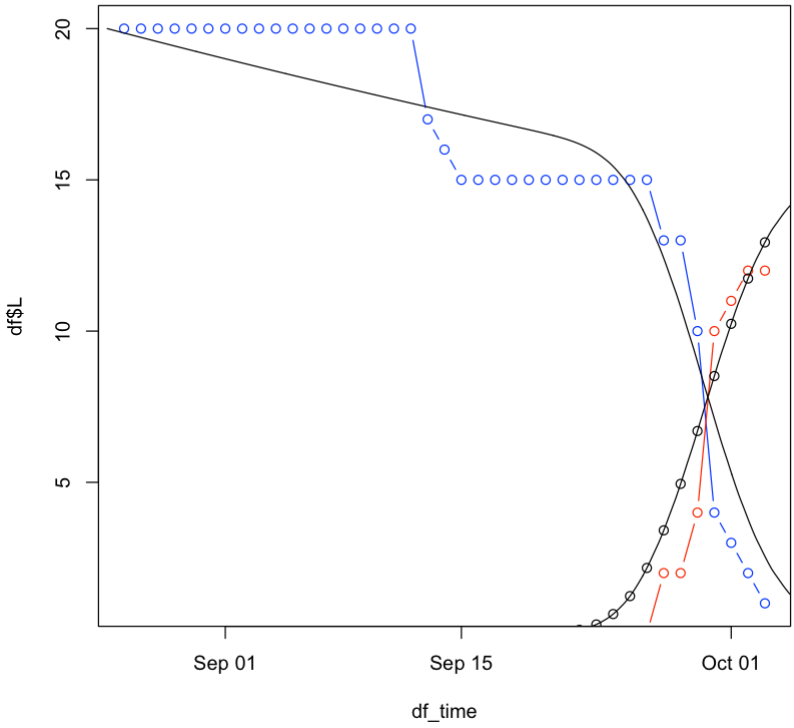
In [64]:
```
best_param <- fit$par
best_sim <- simulate_system(best_param)

plot(df_time,df$L,t='b',col='blue')
lines(df_time,df$P,t='b',col='red')

lines(tm_time, best_sim$larva, t='l')
```

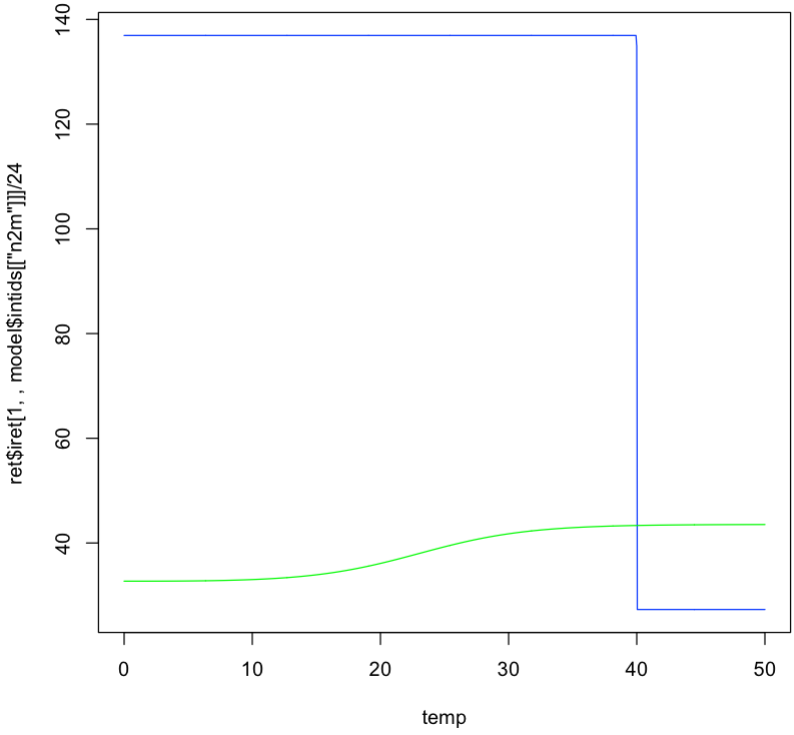
```
lines(tm_time, best_sim$pupa, t='l')

best_sim_at_obs <- best_sim[tm_time %in% df_time, ]

lines(df_time, best_sim_at_obs$pupa, t='p')
```



In [55]:

```
# This works when the standard deviation is 1% of the mean
best_param <- c(-10.0, 6.0, 50.0, 6.0, 655.094187, 2631.398751, 26.608980, -6.0, -3.6

best_sim <- simulate_system(best_param)
best_sim_at_obs <- best_sim[tm_time %in% df_time, ]

plot(df_time,df$L,t='b',col='blue')
lines(df_time,df$P,t='b',col='red')
lines(tm_time, best_sim$larva, t='l')
lines(tm_time, best_sim$pupa, t='l')
lines(df_time, best_sim_at_obs$pupa, t='p')

temp <- seq(0,50,length.out=1000)

ret <- model$sim(
  ftime = length(temp)+1,
  envir = list(
    temp = temp
  ),
  pr = best_param,
  rep = -1
)

plot(temp, ret$iret[1, , model$intids[["n2m"]]] / 24.0, t='l', col='blue')
lines(temp, ret$iret[1, , model$intids[["d2m"]]] / 24.0, t='l', col='green')
```





I hope you found this tutorial informative for getting familiar with the PopJSON representation. You can now explore the *Aedes aegypti* model we have prepared for the CSVD Nicosia2025 Workshop using PopJSON.

In [7]:
```
model <- prepare("models/model_aegypti")
```

```
[1] "Model translated to models/model_aegypti.c"
[2] "Model file created: models/model_aegypti.dylib"
Loaded model with: 7 pops, 59 pars, 34 ints, 7 envs.
```

In [8]:
```
print(model$popnames)
print(model$parnames)
print(model$param)
```

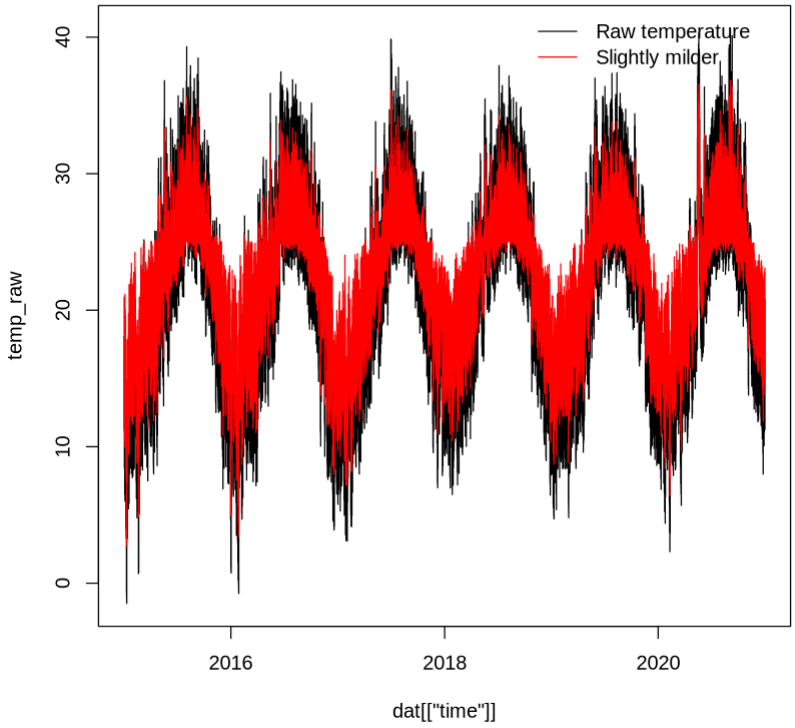
```
[1] "pop_E"   "pop_Eh"  "pop_L"   "pop_P"   "pop_Afj" "pop_Af"  "pop_Afg"
 [1] "par_E_death_L"       "par_E_death_l"       "par_E_death_R"
 [4] "par_E_death_r"       "par_E_death_X"       "par_E_death_str"
 [7] "par_E_death_dsc"     "par_E_dev_R"         "par_E_dev_r"
[10] "par_E_dev_min"       "par_E_dev_max"       "par_E_to_L"
[13] "par_Eh_dev"          "par_Eh_bs_thr"       "par_Eh_T_thr"
[16] "par_L_to_P"          "par_L_death_L"       "par_L_death_l"
[19] "par_L_death_R"       "par_L_death_r"       "par_L_death_X"
[22] "par_L_death_str"     "par_L_dev_min"       "par_L_dev_max"
[25] "par_L_dev_a"         "par_L_fdens"         "par_L_fdens_str"
[28] "par_L_fdens_mort"    "par_L_fdens_dev"     "par_P_death_L"
[31] "par_P_death_l"       "par_P_death_R"       "par_P_death_r"
[34] "par_P_death_X"       "par_P_death_str"     "par_P_dev_min"
[37] "par_P_dev_max"       "par_P_dev_a"         "par_P_to_A"
[40] "par_A_death_str"     "par_female"          "par_bs_prec"
[43] "par_bs_pdens"        "par_bs_nevap"        "par_Khum"
[46] "par_kvec"            "par_F4_M"            "par_F4_a"
[49] "par_F4_b"            "par_gc_X"            "par_gc_M"
[52] "par_gc_a"            "par_A_life_L"        "par_A_life_l"
[55] "par_A_life_R"        "par_A_life_r"        "par_A_life_H"
[58] "par_factor_size"     "par_factor_size_off"
 [1]    10.0000000    -0.8879500    25.0000000    -0.1466167   400.0000000
 [6]     1.0000000    10.0000000    -6.0017909    -1.8236740    16.5313762
[11] 12877.1739646     1.0000000   240.0000000     0.5000000    15.0000000
[16]     0.9000000    12.0000000     0.6000000    22.0000000     0.9295488
[21]  1000.0000000     0.5000000     0.0000000  2051.5571700     0.1000000
[26]   300.0000000    -4.8000000     1.0000000     0.1000000    17.0000000
[31]     0.6000000    17.0000000     0.9295488  1000.0000000     0.5000000
[36]    36.0000000  2051.5571700     0.1500000     0.9000000     0.5000000
[41]     0.5000000     0.0001000     0.0000010    20.0000000   100.0000000
[46]   100.0000000    26.0000000    -0.5000000    45.0000000  1700.0000000
[51]     1.0000000     0.1000000    16.5000000     1.0000000    15.0000000
[56]    -1.2000000   870.0000000     0.0010000     0.1000000
```

In [10]:

```
dat <- read.csv("https://veclim.com/files/workshops/September2025/clim/MissionI/clim_
```

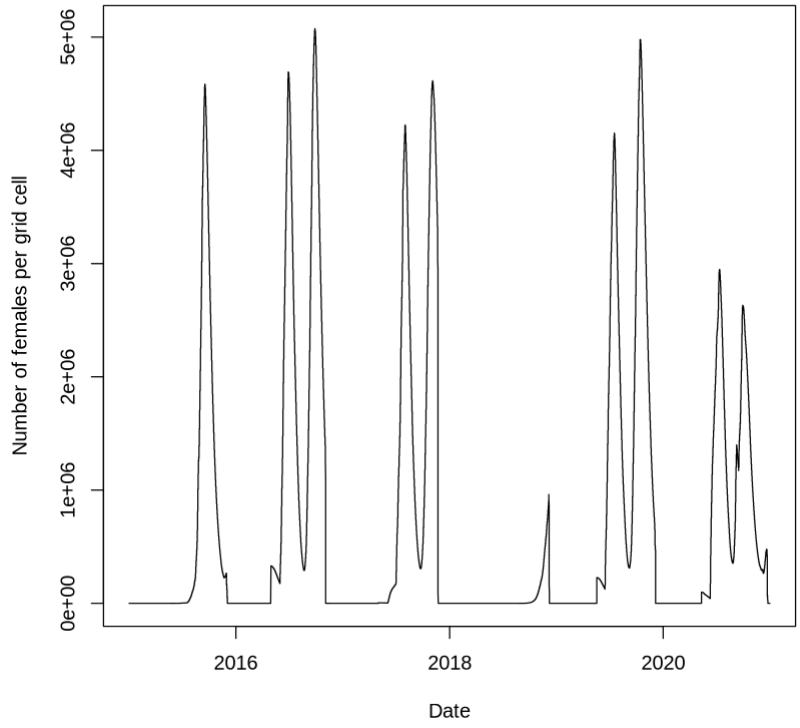
In [24]:

```
source("aux_micro.R")

dat[['time']] <- as.Date(dat[['time']])

temp_raw <- dat[['temp']]
temp_mild <- micro_optimum(temp_raw,2.5)

plot(dat[['time']],temp_raw,t='l',col='black')
lines(dat[['time']],temp_mild,t='l',col='red')
legend("topright",
       legend = c("Raw temperature", "Slightly milder"),
       col = c("black", "red"),
       lty = 1, bty = "n")
```



In [28]:
```r
pr <- c(10,-0.88795,25,-0.146617,400,1,10,-6.00179,-1.82367,16.5314,12877.2,1,240.108

sim <- model$sim(
    ftime = length(dat[['time']]),
    envir = list(
                 temp = temp_mild,
                 prec = dat[['prec']],
                 evap = dat[['evap']],
                 rehum = dat[['rhum']],
                 pdens = dat[['popd']],
                 experiment = c(0),
                 configure = c(1)
                ),
    pr = pr,
    y0 = list(pop_Eh = 1000.0)
    )
```

In [31]:
```r
plot(dat[['time']], sim$ret[1, , model$popids[['pop_Af']]], t='l', xlab="Date", ylab=
```


Number of females per grid cell
0e+00
1e+06
2e+06
3e+06
4e+06
5e+06
2016
2018
2020
Date

In [ ]: